

CO-DESIGN RESEARCH AGENDA

Four ways to make the retention budget an explicit cross-layer resource

1. Retention-aware eviction

suspect state



pin / protect



preserve evidence

Trade bounded memory for replayable diagnostic evidence.

2. Failure-aware scheduling

attribution signal



scheduler



deprioritize risk

Feed reliability evidence back into placement and execution policy.

3. Budgeted observability

SLO / budget



trace fidelity



dynamic capture

Negotiate diagnostic fidelity against serving cost at runtime.

4. Infrastructure-triggered diagnostics

latency / memory



early warning



start diagnosis

Use serving anomalies as triggers before task-level failure surfaces.

All four are research proposals grounded in surveyed components; none is claimed as an implemented system.